

MEMORANDUM

PFAS Contamination of the City of Miami Beach Drinking Water Supply

TO:	Ricardo J. Dopico, Esq, City Attorney, City of Miami Beach Commissioner Tanya Bhatt, City of Miami Beach
FROM:	Gregory J. Rigano, Esq. — Resident, Miami Beach, Florida Attorney, www.RiganoLLC.com
DATE:	May 20, 2026
RE:	High Levels of PFAS ("Forever Chemicals") Contamination in the Drinking Water Supplied to the City of Miami Beach advertised as safe and excellent to children

I. The Issue + Solution, In Plain English

What this memorandum says, in one paragraph

The drinking water that Miami-Dade County sells to the City of Miami Beach contains industrial "forever chemicals" called PFOA and PFOS ("PFAS") at levels exponentially higher than what the EPA has determined to be safe. Both Miami-Dade and the City of Miami Beach advertise the drinking water as safe and excellent with children smiling drinking it to the residents of Miami Beach. Specifically, the PFAS levels are approximately 7 to 10 times the legal limit set by the U.S. EPA in 2024, and approximately 1,900 to 7,500 times the safety level the EPA published in 2022. While the EPA has granted municipalities 5 years to come in compliance with the legal limit (2031), the EPA states that these forever chemicals at these levels are cancer causing, harm development of children and pregnant women, cause thyroid issues, ulcerative colitis, amongst other serious medical conditions. The EPA certainly does not state that these levels of PFAS in drinking water should be advertised to children as safe and excellent. The City of Miami Beach must immediately install certified point-of-use or point-of-entry PFAS-removal filtration at every public school, child-care facility, and public-access drinking fountain within the City of Miami Beach, prioritizing facilities serving children and pregnant women. PFAS filtration is commercially available as a tried-and-true method of removing 99% of PFAS from drinking water.

The source of these chemicals is firefighting foam (AFFF) used at Miami International Airport (a known Superfund site) and Miami-area fire training facilities. The PFAS contamination flows through the Biscayne Aquifer from west to east, into the production wells located at and east of the airport,

and into the finished water served to City of Miami Beach customers — **including children at every public school**. The water can be cleaned. Reverse osmosis and anion exchange — both recognized by the EPA as Best Available Technologies — remove more than 99% of these chemicals. Notably, it is established that long-term exposure to PFOA and PFOS at concentrations far below those measured in Miami Beach drinking water can result in kidney cancer, testicular cancer, immune suppression, thyroid disease, pregnancy-induced hypertension, high cholesterol, hormonal issues and many other conditions. Treatment by Miami-Dade County is not scheduled to be in place until 2031. In the meantime, the City of Miami Beach has the unilateral authority to (a) warn residents in plain language, (b) install certified filtration in every school and child-care facility, (c) test the water it delivers and (d) most importantly STOP ADVERTISING THE DRINKING WATER AS SAFE AND EXCELLENT TO CHILDREN.

Five facts that drive everything else

- **Fact 1 — The numbers.** The 2023 City of Miami Beach Annual Water Quality Report (the official report distributed to every utility customer) shows PFOS at 38 parts per trillion (ppt) and PFOA at 30 ppt.
- **Fact 2 — The legal limit.** On April 10, 2024, EPA finalized the first nationwide legal limit for PFOA and PFOS in drinking water at 4 ppt each. The PFOS in Miami Beach water is approximately 9.5 times the legal limit; the PFOA is approximately 7.5 times the legal limit.
- **Fact 3 — The safety level.** EPA's 2022 Lifetime Health Advisory levels — the levels designed to protect children, pregnant women, and people drinking the water for a lifetime — are 0.004 ppt for PFOA and 0.02 ppt for PFOS. Miami Beach water exceeds these by approximately 7,500× and 1,900×, respectively.
- **Fact 4 — The source.** The Northwest Wellfield (west of Miami International Airport) tests at 1–2.3 ppt PFOA and 0.4–2.0 ppt PFOS — about ten times less than the water being served. The wells located at and east of the airport are the contaminated wells. Independent peer-reviewed research has identified the chemical fingerprint of airport firefighting foam in the contamination. Only wells to the WEST of MIA should be used, everything else should be shut down.
- **Fact 5 — The fix.** EPA-recognized treatment technologies — reverse osmosis, ion exchange — remove 99%+ of PFAS. They are commercially available, are already deployed in Hialeah and North Miami Beach, and are certified for residential point-of-use installation under NSF/ANSI Standards 53, 58, and 401. This won't come online until 2031 – if no one does anything children and pregnant women continue to consume cancer causing water for another 5 years (in addition to what they have already consumed). PFAS chemicals are called forever chemicals because they generally do not break down and accumulate in the human body and food chain.

II. The Source: Miami International Airport, AFFF Firefighting Foam, and the Biscayne Aquifer Superfund Sites

A. What AFFF Is and Why It Matters

AFFF ("aqueous film-forming foam") is the white firefighting foam used since the late 1960s to put out jet-fuel fires. The U.S. military, every commercial-service airport, and most municipal fire-training facilities have used it. Until very recently, every formulation of AFFF certified to the U.S. military specification (MIL-PRF-24385) — the specification the FAA required for Part 139 commercial-service airports until the FAA's 2018 reauthorization began the transition — contained PFOA, PFOS, and other PFAS. When AFFF is sprayed during a fire, an emergency, a training burn, or a system test, it drains into the soil and groundwater. PFAS does not break down. It accumulates.



Aircraft fire response at Miami International Airport (Red Air MD-82, February 21, 2022). The white foam visible on and around the aircraft is AFFF — the firefighting foam containing PFOA, PFOS, and other PFAS chemicals. Source: news media coverage of the incident.

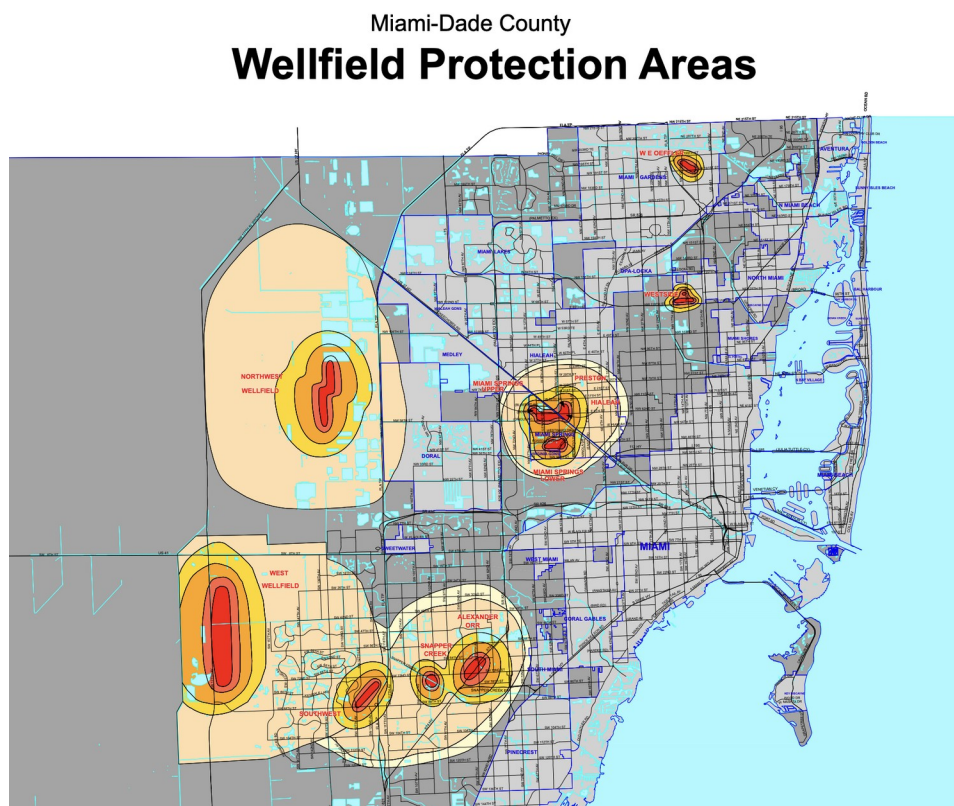
B. Miami-Dade County Owns Both the Airport and the Drinking Water

Miami-Dade County owns and operates Miami International Airport ("MIA") through its Aviation Department and owns and operates the public drinking-water system through the Miami-Dade Water and Sewer Department ("MDWASD"). MIA has been an active commercial-service airport since 1928. AFFF has been used at MIA continuously since at least the early 1970s for fire response, training, and equipment testing. The result is that the same county that contaminated the groundwater with AFFF also pumps the contaminated groundwater out of the ground and sells it to the residents of the City of Miami Beach. There is no intermediary.

C. The Wellfields and the Wellfield Map

The Miami-Dade wellfields that supply the City of Miami Beach — the Hialeah, Miami Springs (Upper and Lower), Preston, and Alexander Orr wellfields — are located at and immediately east of MIA. The

Northwest Wellfield is located west of MIA. The map below, published by Miami-Dade County, shows the wellfield protection areas. Groundwater in the Biscayne Aquifer flows generally from west to east. The Northwest Wellfield therefore sits upgradient of MIA; the wells that supply Miami Beach sit at or downgradient of MIA. This explains why the Northwest Wellfield tests at 1–2 ppt PFOA/PFOS while the finished water served to Miami Beach tests at 30–38 ppt — approximately ten times higher.



Miami-Dade County Wellfield Protection Areas map. The Northwest Wellfield (west, upgradient of MIA) is comparatively clean. The wellfields at and east of MIA — Hialeah, Miami Springs, Preston, Alexander Orr — supply the contaminated water served to City of Miami Beach residents. Source: Miami-Dade County Department of Environmental Resources Management (DERM).

D. FDEP's Site Assessments Confirmed AFFF as a Source — but Skipped MIA

Beginning in approximately 2019, the Florida Department of Environmental Protection ("FDEP") commenced PFAS site assessments at Florida fire-training facilities suspected of using AFFF. Four such facilities are in the Miami area:

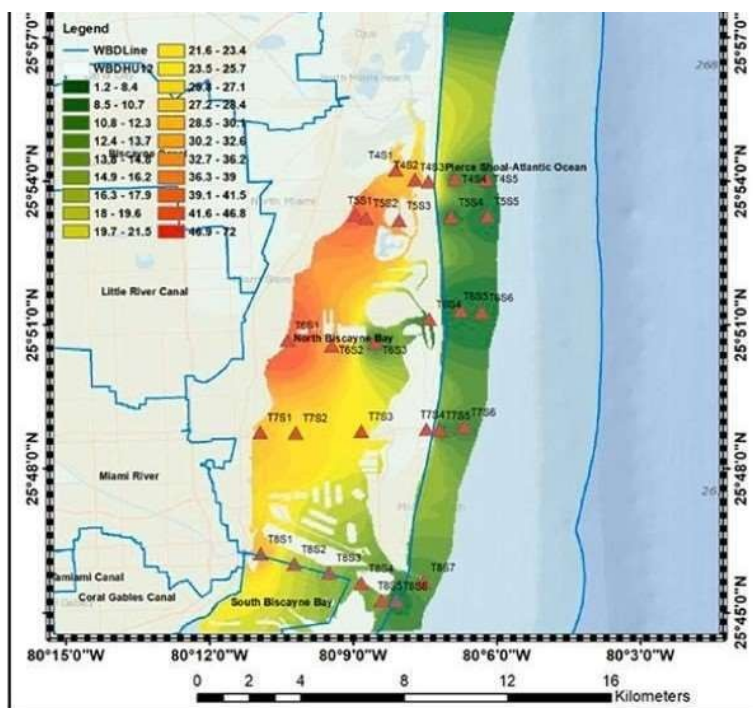
- Miami-Dade Academy of Fire Science
- City of Miami Fire Rescue Training Center
- Miami-Dade Fire Rescue Training Center
- City of Hialeah Training Facility

FDEP's published assessments confirmed AFFF use at three of the four facilities and confirmed PFAS detections in groundwater exceeding EPA standards at all four facilities. FDEP also investigated the Former

Upsilon-Davis Plating Facility (a separate Superfund site) and confirmed PFAS detections in groundwater exceeding EPA standards. On the public record reviewed by the undersigned, FDEP has not conducted a formal PFAS site assessment of MIA itself, despite the documented operational history of AFFF use at the airport. The absence of a published MIA-specific FDEP PFAS site assessment is a material data gap in the regulatory record.

E. Independent Peer-Reviewed Research Identified the Chemical Fingerprint of Airport AFFF

A 2024 peer-reviewed study by Ogunbiyi, Massenat, and Quinete (Florida International University Institute of Environment), published in Science of the Total Environment (DOI 10.1016/j.scitotenv.2023.168413), identified 6:2 fluorotelomer sulfonate ("6:2 FTS") — a chemical signature characteristic of AFFF formulations used at airports and military bases — as the dominant PFAS detected in the Miami River. The Miami River flows past Miami International Airport before discharging into Biscayne Bay. The study reported PFAS "hot spots" at the discharge points of the Miami River, Little River, and Biscayne Canal, with concentrations decreasing as samples moved further offshore. A 2021 peer-reviewed FIU study (Vega Millán et al., Quinete) sampled 21 South Florida tap-water locations and reported the highest single tap-water PFAS concentration in the entire study area — 242 ppt total PFAS — at Grapeland Heights, the residential neighborhood immediately adjacent to MIA. The second-highest concentration (124 ppt) was at Dania Beach, immediately adjacent to Fort Lauderdale-Hollywood International Airport.



FIU Institute of Environment heat map of PFAS concentrations in Biscayne Bay. Red and orange indicate the highest concentrations; green indicates low concentrations. The hot spots correspond to the discharge points of the Miami River, Little

River, and Biscayne Canal — the canals that drain westward through the airport area to the bay. Source: Quinete & Ogunbiyi (FIU), 2023, republished with permission.

III. The Numbers

A. Reported Concentrations in the Drinking Water

The City of Miami Beach ("CMB") purchases all its drinking water from MDWASD; CMB does not operate its own water-supply wells, treatment plants, or PFAS-removal infrastructure. PFAS sampling and analysis for the water supplied to CMB is performed by MDWASD and reported in the MDWASD Annual Consumer Confidence Report ("CCR") and in the City of Miami Beach Annual Water Quality Report. CMB redistributes the same finished water to approximately 90,000 daily customers.

Compound	City of Miami Beach (2023 CCR)	Miami-Dade WASD (2023 CCR)	EPA 2022 Lifetime Health Advisory	EPA 2024 Final MCL
PFOS	38 ppt	34 ppt	0.02 ppt	4 ppt
PFOA	30 ppt	29 ppt	0.004 ppt	4 ppt

B. Magnitude of the Exceedance

Compound	Times over EPA Final MCL (4 ppt)	Times over EPA 2022 Health Advisory
PFOS in Miami Beach water (38 ppt)	≈ 9.5× higher	≈ 1,900× higher
PFOA in Miami Beach water (30 ppt)	≈ 7.5× higher	≈ 7,500× higher

The Maximum Contaminant Level Goal (MCLG) — the health-based goal set by EPA — is zero for both PFOA and PFOS, because EPA has determined that PFOA and PFOS are likely human carcinogens for which no safe threshold of exposure can be identified. Comparable concentrations have been published in the Miami-Dade and City of Miami Beach Annual Water Quality Reports continuously since voluntary PFAS testing began in 2019.

C. Northwest Wellfield Comparison — The Order-of-Magnitude Difference

I, Gregory Rigano, obtained from MDWASD, at my own expense, the most recent PFAS sampling data for the Northwest Wellfield (located west of MIA in unincorporated Miami-Dade County). That data shows the following:

Source / Location	PFOA Range	PFOS Range
Northwest Wellfield (west of MIA, upgradient)	1.0 – 2.3 ppt	0.4 – 2.0 ppt
Finished water served to Miami Beach (MIA-area wellfields)	29 – 30 ppt	34 – 38 ppt

The Northwest Wellfield concentrations are approximately one order of magnitude (10×) lower than the concentrations reported in the finished water that is delivered to City of Miami Beach residents. The most parsimonious explanation for this difference, consistent with the regional hydrogeology and with the documented AFFF source areas, is that the Northwest Wellfield is upgradient (west) of the dominant PFAS sources — MIA and the Miami-area fire training facilities — while the wellfields supplying the City of Miami Beach (Hialeah, Miami Springs, Preston, Alexander Orr) are at or downgradient of those sources.

IV. The Regulatory Framework

A. EPA's April 2024 Final Rule

On April 10, 2024, EPA promulgated the first legally enforceable national drinking water standard for six PFAS compounds. The rule establishes the following Maximum Contaminant Levels ("MCLs") and Maximum Contaminant Level Goals ("MCLGs"):

Compound	MCLG (Health-Based Goal)	Final MCL (Enforceable)	2022 Lifetime Health Advisory
PFOA	Zero	4.0 ppt (ng/L)	0.004 ppt
PFOS	Zero	4.0 ppt (ng/L)	0.02 ppt
PFHxS	10 ppt	10 ppt	—
PFNA	10 ppt	10 ppt	—
HFPO-DA (GenX)	10 ppt	10 ppt	—

EPA additionally promulgated a Hazard Index of 1 (unitless) applicable to mixtures of two or more of PFHxS, PFNA, HFPO-DA, and PFBS. Initial monitoring is required to be completed by 2027. The original compliance deadline by which utilities must implement treatment to meet the MCLs was 2029. On May 14, 2025, EPA announced that it intends to retain the MCLs for PFOA and PFOS at 4.0 ppt and to extend the compliance deadline to 2031, while proposing to rescind the MCLs and reconsider the regulatory determinations for PFHxS, PFNA, HFPO-DA, and the Hazard Index. As of the date of this memorandum, the PFOA and PFOS MCLs of 4.0 ppt remain in force.

B. EPA's 2022 Lifetime Health Advisories

On June 15, 2022, EPA issued updated Lifetime Drinking Water Health Advisories of 0.004 ppt for PFOA and 0.02 ppt for PFOS. EPA's published rationale states that these levels identify the concentrations "to protect all people, including sensitive populations and life stages, from adverse health effects resulting from exposure throughout their lives to these PFAS in drinking water." Health Advisories are non-enforceable and non-regulatory in nature but represent EPA's expert assessment of the concentrations below which adverse health effects are not anticipated.

V. Why Long-Term, Low-Dose Exposure Matters: The C8 Science Panel

A. What Happened in the Mid-Ohio Valley

From 1951 through approximately 2002, the DuPont Washington Works chemical plant on the Ohio River used approximately 1.7 million pounds of PFOA in the manufacture of Teflon. PFOA escaped to the air, the soil, and the Ohio River, and into the drinking-water systems of approximately six water districts in West Virginia and Ohio. In 2001, attorneys representing residents of those water districts filed a class-action lawsuit against DuPont. The 2005 settlement of that lawsuit (*Leach v. E. I. du Pont de Nemours & Co.*) created and funded the C8 Health Project — the largest population-level human-health study of PFOA exposure ever conducted, and the largest-ever study of any single chemical.

B. The Numbers Behind the C8 Findings

- Approximately 69,000 PFOA-exposed residents enrolled in the study and provided blood serum, medical history, and environmental-exposure information.
- The C8 Science Panel — three independent epidemiologists agreed upon by both DuPont and the plaintiffs — was charged with reviewing the data and issuing "Probable Link" determinations identifying which medical conditions had a more-likely-than-not causal connection to PFOA exposure.
- The Panel worked from 2005 through 2012 (approximately seven years) and reviewed approximately 2,500 validated cancer cases across 21 different cancer types and a much larger number of non-cancer health outcomes.

C. The C8 "Probable Link" Determinations (2012)

The C8 Science Panel issued formal Probable Link determinations identifying the following six medical conditions as having a probable causal link to PFOA exposure in the Mid-Ohio Valley population:

1. Kidney cancer
2. Testicular cancer
3. Thyroid disease
4. Ulcerative colitis (an autoimmune inflammatory bowel disease)

5. Pregnancy-induced hypertension (including preeclampsia)
6. Diagnosed high cholesterol (hypercholesterolemia)

These six Probable Link determinations form a substantial portion of the underlying scientific record on which EPA based (a) the 2022 Lifetime Health Advisories of 0.004 ppt for PFOA and 0.02 ppt for PFOS, (b) the 2024 final MCL of 4.0 ppt for both compounds, and (c) the 2024 final MCLG of zero for both compounds.

D. The Critical Point: Bioaccumulation Means Long-Term Exposure Produces High Internal Doses Even at Low Drinking-Water Concentrations (CMB is not “low”)

Why long-term matters

PFOA and PFOS bioaccumulate in the human body — once consumed, they do not leave easily. The biological half-life of PFOA in humans is approximately 3.5 years (geometric mean; 95% CI 3.0–4.1 years per Olsen et al. 2007), and the half-life of PFOS is approximately 4.8 years (geometric mean; 95% CI 4.0–5.8 years per Olsen et al. 2007). At steady state, EPA exposure modeling and peer-reviewed pharmacokinetic studies indicate that serum PFOA concentrations in chronically exposed populations are approximately 100-fold higher than the drinking-water concentration consumed (i.e., a person consuming water at 30 ppt PFOA every day for years can be expected to develop a serum PFOA concentration on the order of 3,000 ppt, or 3 ng/mL, at steady state). For children, who consume more water per pound of body weight than adults, the steady-state internal dose is correspondingly higher.

This means that a person who consumes contaminated water at 30 ppt PFOA every day will, over years, accumulate PFOA in their blood at concentrations many times higher than the drinking-water concentration. The same biological mechanism that produced cancers and disease over decades of low-dose exposure as determined by the EPA operates on every Miami Beach resident — including every Miami Beach child — drinking the same contaminated water from the same tap, every day, for years. Everyday matters – waiting until 2031 is unacceptable.

VI. The Documented Health Effects of PFAS

A. EPA's Documented Health Effects

EPA's published guidance identifies the following adverse outcomes associated with PFAS exposure at levels found in drinking water:

- Reproductive effects, including decreased fertility and increased high blood pressure in pregnant women.

- Developmental effects or delays in children, including low birth weight, accelerated puberty, bone variations, and behavioral changes.
- Increased risk of certain cancers, including prostate, kidney, and testicular cancers.
- Reduced ability of the immune system to fight infections, including reduced vaccine response.
- Interference with the body's natural hormones.
- Increased cholesterol levels and increased risk of obesity.
- Cardiovascular effects.
- Liver effects.

EPA further states that children are more sensitive to the harmful effects of PFAS than adults because they are still developing, and they are exposed to more PFAS per pound of body weight than adults because they consume more water, food, and air per pound of body weight.

B. Vaccine-Response and Immune Effects (Grandjean et al.)

Grandjean et al. (2012) reported in the Journal of the American Medical Association (JAMA, vol. 307, no. 4, pp. 391–397) on a prospective birth-cohort study in the Faroe Islands. The study found that a doubling of serum PFOS concentration at birth was associated with an approximately 40 percent reduction in serum diphtheria-antibody concentration at age 5, and that doubling serum PFOS or PFOA at age 5 was associated with a 2.4- to 4.2-fold increase in the likelihood of antibody concentrations falling below the protective level for tetanus and diphtheria at age 7. Grandjean et al. (2017, Journal of Immunotoxicology) and additional Faroe Islands cohort follow-ups have replicated these findings. Reduced antibody response to vaccines is one of the most sensitive adverse-health endpoints used by EPA in deriving the 2024 MCLs.

C. Endocrine Disruption / Androgen Axis

Multiple peer-reviewed studies have reported that PFAS, including PFOA and PFOS, act as endocrine-disrupting compounds capable of (a) inhibiting steroidogenesis (including inhibition of CYP11A1 and 17 β -HSD), (b) interfering with the hypothalamic-pituitary-gonadal axis, (c) binding to and altering the activity of androgen receptors and sex-hormone-binding globulin, (d) inducing Leydig-cell hyperplasia, and (e) producing epigenetic changes in genes related to steroidogenesis and hormone regulation. Reported epidemiologic findings include negative associations between serum PFAS concentrations and serum testosterone in adolescents and young adults (Tsai et al., 2015), associations between PFAS exposure and reduced sperm quality, and associations between PFAS exposure and delayed sexual maturation in boys (Rappazzo et al., 2017; Bach et al., 2016).

D. Bioaccumulation and Persistence

PFOA, PFOS, and other long-chain PFAS are persistent in both the environment and the human body. The biological half-life of PFOA in humans has been reported in peer-reviewed literature at approximately 2.7–3.5 years, and the half-life of PFOS at approximately 4.8–5.4 years. Continuous exposure through drinking

water at the levels reported in the Annual Water Quality Reports for the City of Miami Beach therefore results in continuous bioaccumulation in the bodies of consumers.

VII. The City's Own Communications

A. The Annual Water Quality Report

The Annual Water Quality Reports issued by MDWASD and by the City of Miami Beach Public Works Department for the years 2019 through 2023 (the most recent year published) describe the drinking water as follows:

- "The Water Quality Report is designed to inform you about the excellent water the City of Miami Beach delivers to you every day."
- "Our number one goal is to provide you and your family a safe and dependable supply of drinking water."
- "To ensure the safety of your water, [the utility] routinely monitors for contaminants in your drinking water according to federal, state, and local laws, rules and regulations."
- The reports include photographs of children drinking the tap water (reproduced below).



For Customers with Special Health Concerns

Some people may be more vulnerable to contaminants in drinking water than the general population. Immuno-compromised persons such as persons with cancer undergoing chemotherapy, persons who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and infants can be particularly at risk from infections. These people should seek advice about drinking water from their health care providers. EPA/CDC guidelines on appropriate means to lessen the risk of infection by *Cryptosporidium* and other microbiological contaminants are available from the Safe Drinking Water Hotline at 1-800-426-4791.

Para los Consumidores con Preocupaciones Especiales de Salud

Es posible que algunas personas sean más vulnerables a los contaminantes hallados en el agua potable que la población en general. Las personas con sistemas inmunológicos afectados por diversas razones como los pacientes de cáncer que reciben tratamiento de quimioterapia, personas que han recibido un trasplante de órgano, personas con VIH o SIDA u otra enfermedad del sistema inmunológico, algunas personas mayores y bebés, pueden tener mayor riesgo de infección. Estas personas deben consultar el uso de agua potable con su médico. Las pautas de la EPA/CDC sobre las maneras apropiadas de reducir el riesgo de infección por exposición a *criptosporidio* y otros contaminantes microbiológicos están disponibles a través de Línea Directa del Agua Potable Segura en el 1-800-426-4791.

Image reproduced from the City of Miami Beach Annual Water Quality Report (and from the substantively identical Miami-Dade WASD Annual CCR). The report describes the water as "excellent" and "safe" while disclosing PFOS and PFOA concentrations approximately 9.5× and 7.5× the EPA Final MCL, respectively, and approximately 1,900× and 7,500× the EPA 2022 Lifetime Health Advisory. Source: City of Miami Beach Annual Water Quality Report, 2023.

B. MDWASD's and CMB Water Stated Position

MDWASD has tested for PFAS since 2019 and has known for years that its finished water exceeds the EPA Health Advisory levels, and during that period MDWASD and the City of Miami Beach have published Annual Water Quality Reports characterizing the water as "excellent" and "safe" and depicting children

consuming the water. In correspondence dated February 20, 2026 and March 12, 2026, MDWASD Director Jay J. Fink, P.E., reiterated that Miami-Dade drinking water "continues to comply with all Environmental Protection Agency (EPA) National Primary Drinking Water Regulations." Director Fink did not address any of the specific factual items raised in the underlying correspondence, did not name Miami International Airport — owned and operated by the same county — as a source of the PFAS contamination, and did not commit to any change in the public-facing characterization of the finished water.

EPA's April 10, 2024 Final Rule sets the legally enforceable Maximum Contaminant Level for PFOA and PFOS at 4.0 ppt and the corresponding MCL Goals at zero; the City of Miami Beach finished water at 30 ppt PFOA and 38 ppt PFOS exceeds the MCL by approximately 7.5 and 9.5 times, and exceeds the 2022 EPA Lifetime Health Advisories (0.004 ppt PFOA; 0.02 ppt PFOS) by approximately 7,500 and 1,900 times. EPA's published rationale for the Lifetime Health Advisories states that those levels protect "all people, including sensitive populations and life stages," from adverse health effects. The EPA-granted compliance grace period (originally 2029, extended in May 2025 to 2031) authorizes continued distribution of finished water exceeding the MCL while treatment infrastructure is being constructed; EPA has not, in any regulation as stated by Fink, authorized any utility to characterize finished water exceeding the MCL as "excellent" or "safe" in public-facing communications during that grace period. In correspondence dated March 13, 2026, I specifically requested that MDWASD identify the EPA regulation that authorizes the "excellent" and "safe" characterization of MCL-exceeding finished water; no response identifying any such regulation has been received. The EPA has characterized the PFAS levels as cancer causing and dangerous — the same PFAS levels City of Miami Beach and Miami Dade promote as safe and excellent.

Notably, Miami-Dade County is itself a plaintiff in *In re Aqueous Film-Forming Foams (AFFF) Products Liability Litigation*, MDL No. 2873, in which the County is seeking damages reported at approximately \$199 million on the documented theory that the AFFF manufacturers' PFAS chemicals contaminated the County's drinking-water supply. The City of Miami Beach is not the operator of the water system but the wholesale customer of MDWASD; Director Fink is a Miami-Dade County employee, and his stated positions are advanced on behalf of the County in its three concurrent roles — operator of the airport (the documented PFAS source), operator of the water utility (the seller of the contaminated finished water), and plaintiff against the AFFF manufacturers. The City of Miami Beach and its approximately 90,000 daily water customers occupy a different position: they consume the finished water and bear the public-health and economic exposure that flows from any failure to disclose the contamination, recommend filtration, or change the wellfield-supply mix. MDWASD has not issued any public-health advisory disclosing the measured concentrations, the EPA Final MCLs, the Lifetime Health Advisories, or the EPA-published health effects; has not recommended NSF/ANSI 53, 58, or 401-certified residential point-of-use filtration (commercially available at retail for \$200–\$500 per unit, certified to remove more than 99 percent of PFAS); has not maximized supply from the upgradient Northwest Wellfield (which tests at 1.0–2.3 ppt PFOA, approximately one order of magnitude below the finished-water concentrations); and has not removed the "excellent" and "safe" characterization or the photographs of children consuming the water from the Annual Water Quality Reports. The continued public-facing characterization of finished water

exceeding the EPA Final MCL by 7.5 to 9.5 times as "excellent" and "safe" — for years, by a County entity that is simultaneously suing the chemical manufacturers for contaminating that same water — is the documented record on which the City of Miami Beach must now make its own determination of how to protect its residents.

Additionally I met with Amy Knowles, Chief Resilience Officer, City of Miami Beach and John Norris, Public Works Director, Miami Beach in December 2025 and presented them with the dangers of PFAS according to the EPA, the public must be warned, the language advertising and promoting the water as safe and excellent is incorrect and should be fixed as well as the sources of contamination and how the Northwest Wellfield is significantly cleaner. The Miami Beach Water Quality Report that was issued after this meeting did include updated language but still features that the water is safe and excellent as well as children playing in the water and drinking it. **"The data confirms that the city's drinking water quality continues to be excellent — not just safe for our more than 80,000 residents but also for the millions of visitors who travel to Miami Beach each year."** The report also features the following children showing that the water is safe and excellent for children.



Notably, the 2023 City of Miami Beach Water Quality Reports, issued in 2024, were published two years after EPA's 2022 lifetime health advisories for PFOA and PFOS—0.004 ppt and 0.02 ppt, respectively—and its zero MCLG for both compounds, yet prior reports prominently featured images of children smiling and drinking the water.



- The City of Miami Beach Annual Water Quality Reports do not currently include any public-health advisory regarding PFAS, do not currently advise consumers of the availability of point-of-use treatment, and do not currently disclose the EPA Final MCL or 2022 Lifetime Health Advisory comparisons.
- PFAS testing has not, on the public record, been conducted at the point of use within the public schools, child-care facilities, or hospitals located within the City of Miami Beach.

VIII. Suggested Action Items for the City of Miami Beach

The following actions are within the unilateral authority of the City of Miami Beach, independent of any action by Miami-Dade County or by EPA:

1. Update the City of Miami Beach Annual Water Quality Report to (a) remove characterizations of the water as "excellent" and "safe" pending PFAS treatment, (b) remove photographs of children drinking the tap water, (c) include the EPA Final MCL of 4.0 ppt for PFOA and PFOS, the 2022 Lifetime Health Advisories of 0.004 ppt and 0.02 ppt, the MCLG of zero, and the comparison to the City's reported levels, and (d) include EPA's published list of health effects associated with PFAS exposure.
2. Issue a public-health advisory to City residents using the EPA's published language regarding PFAS health effects, the populations most at risk (children, pregnant women, immunocompromised individuals), and the availability of NSF/ANSI 53, 58, and 401-certified point-of-use filtration.

3. Conduct independent PFAS testing of the finished water at the point of delivery into the City of Miami Beach distribution system, and at representative points of use within the distribution system, to verify the MDWASD-reported concentrations.
4. **Conduct PFAS testing of the drinking water served at every public school, child-care facility, public park drinking fountain, hospital, and senior-living facility located within the City of Miami Beach.**
5. **Install certified point-of-use or point-of-entry PFAS-removal filtration at every public school, child-care facility, and public-access drinking fountain within the City of Miami Beach, prioritizing facilities serving children and pregnant women - immediately.**
6. Formally request that MDWASD (a) prioritize maximizing finished-water supply from the Northwest Wellfield and other wellfields located west of MIA, (b) reduce reliance on the wellfields located at or east of MIA pending source remediation, and (c) publish PFAS testing results for post-treatment finished water at the Hialeah RO Plant.
7. Formally request that FDEP and EPA conduct a PFAS site assessment of MIA equivalent to the assessments completed at the Miami-Dade Academy of Fire Science and the City of Miami and Miami-Dade Fire Rescue Training Centers, and an updated CERCLA review of the Biscayne Aquifer Sites cluster to specifically address PFAS as a contaminant of concern.
8. Formally request that Miami-Dade County publish (a) the timeline for central PFAS treatment-system commissioning at each MDWASD treatment facility, and (b) the proposed allocation of any 3M and other AFFF MDL settlement proceeds to PFAS treatment infrastructure and to interim-period mitigation including point-of-use filtration. Take a seat at the table to understand treatment strategies and well locations to be treated. It is unlikely the wells around the airport can be reduced to safe levels. Only wells to the west can likely be treated to safe levels.
9. Pursue available federal funding for PFAS treatment infrastructure under the Bipartisan Infrastructure Law (which appropriated \$9 billion for PFAS and other emerging-contaminant remediation in drinking water) and under the FAA Reauthorization Act PFAS Replacement Grant Program for airport AFFF transition.
10. Adopt a municipal ordinance requiring all new construction and substantial renovation of City-owned facilities serving children (schools, parks, libraries, recreation facilities) to include PFAS-certified point-of-use or point-of-entry filtration.

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